

Uniform Trace-Class Möbius–Bergman Families with Quantified Order Sensitivity

Hénon Route-A Working Series, C137

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Abstract

We promote a two-branch Möbius transfer operator from one digit pair to the rectangle $(a, b) \in [3, 7/2] \times [6, 7]$. The closed branch images remain separated by at least $1/45$, while the unweighted composition sum on normalized Bergman space is trace class with norm at most $89/16$ and is trace-norm Lipschitz in both parameters. Every word has an exact quadratic-surds trace, giving all power traces and a primitive Fredholm product. The non-cyclic same-count words $aaabb$ and $aabab$ have matrix trace gap $a(b-a)^2 \geq 175/8$, so order sensitivity holds uniformly. These are source-dynamical statements, not an arithmetic or target spectral identification.

1 Family-level progress

Let $\mathbb{D} = \{z : |z| < 1\}$ and equip $A^2(\mathbb{D})$ with normalized area, so $e_n(z) = \sqrt{n+1}z^n$ is orthonormal. Freeze

$$\phi_x(z) = \frac{1}{x+z}, \quad \mathcal{L}_{a,b} = C_{\phi_a} + C_{\phi_b}, \quad (a, b) \in \mathcal{R}_* = [3, 7/2] \times [6, 7].$$

The advance over a fixed pair is uniform: no conclusion below depends on choosing one parameter sample.

2 Uniform geometry and nuclearity

The image of $\overline{\mathbb{D}}$ under ϕ_x is the closed disk with center $x/(x^2-1)$ and radius $1/(x^2-1)$. Its real extremes are $1/(x+1)$ and $1/(x-1)$, hence the gap between the two images is

$$g(a, b) = \frac{1}{a+1} - \frac{1}{b-1} \geq \frac{1}{9/2} - \frac{1}{5} = \frac{1}{45}.$$

Theorem 1 (uniform trace class). *For every $(a, b) \in \mathcal{R}_*$, $\mathcal{L}_{a,b}$ is trace class and $\|\mathcal{L}_{a,b}\|_1 \leq 89/16$.*

Proof. The rank-one expansion $C_{\phi_x}f = \sum_{n \geq 0} \langle f, e_n \rangle \sqrt{n+1} \phi_x^n$ is nuclear because $\sup_{\mathbb{D}} |\phi_a| \leq 1/2$ and $\sup_{\mathbb{D}} |\phi_b| \leq 1/5$. Using $\sqrt{n+1} \leq n+1$ gives

$$\|\mathcal{L}_{a,b}\|_1 \leq \sum_{n \geq 0} (n+1)2^{-n} + \sum_{n \geq 0} (n+1)5^{-n} = 4 + \frac{25}{16}.$$

□

Proposition 1 (trace-norm continuity). *For two parameters in $\mathcal{R}_*$,*

$$\|\mathcal{L}_{a,b} - \mathcal{L}_{a',b'}\|_1 \leq 4|a - a'| + \frac{5}{32}|b - b'|.$$

Proof. The resolvent identity gives $|\phi_x - \phi_y| \leq |x - y|/(m-1)^2$ when $x, y \geq m$. Combining the difference of powers with the nuclear series and $\sum_{n \geq 1} (n+1)nr^{n-1} = 2/(1-r)^3$ gives the stated constants at $(m, r) = (3, 1/2)$ and $(6, 1/5)$. □

3 All words and the determinant

Put $M_x = \begin{pmatrix} 0 & 1 \\ 1 & x \end{pmatrix}$. For $M_w = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$, write $t = \text{tr } M_w$, $\delta = (-1)^{|w|}$, and $\Delta = t^2 - 4\delta$. Directly solving the fixed equation gives

$$z_w = \frac{A - D + \sqrt{\Delta}}{2C}, \quad \lambda_w = \frac{t - \sqrt{\Delta}}{t + \sqrt{\Delta}}, \quad \text{Tr } C_{\Phi_w} = \frac{1}{2} + \frac{t}{2\sqrt{\Delta}}.$$

Expansion of $\mathcal{L}_{a,b}^n$ therefore gives every power trace. Regrouping repetitions in the Fredholm logarithm yields

$$\det(I - z\mathcal{L}_{a,b}) = \prod_{[p]} \prod_{k \geq 0} (1 - z^{|p|} \lambda_p^k), \quad |z| < \frac{1}{2}. \quad (1)$$

The left side is entire by Theorem 1; equation (1) states raw-factor convergence only in the displayed disk.

4 Uniform order sensitivity

The words $aaabb$ and $aabab$ have the same symbol population and are not cyclic rotations. Exact multiplication gives

$$\begin{aligned} t_1 &= a^3b^2 + a^3 + 2a^2b + 2ab^2 + 3a + 2b, \\ t_2 &= a^3b^2 + 4a^2b + ab^2 + 3a + 2b, \end{aligned}$$

so

$$t_1 - t_2 = a(b - a)^2 \geq \frac{175}{8}.$$

Also $t_1 \leq 10731/4$. For $F(t) = 1/2 + t/(2\sqrt{t^2 + 4})$, $F'(t) = 2/(t^2 + 4)^{3/2}$. The mean-value theorem therefore gives

$$F(t_1) - F(t_2) \geq \frac{2800}{(10731^2 + 64)^{3/2}} > 0.$$

5 Exact receipt and boundary

At nine rational sentinels, the release reconstructs all words through period ten: $9(2^1 + \dots + 2^{10}) = 18,414$ rooted receipts and $9 \cdot 226 = 2,034$ primitive-parameter receipts. A separate symbolic run performs 18,379 exact checks; replay is byte-identical and 40 repaired-hash plus one stale-hash mutations are rejected. The grid audits the implementation and is not the proof of uniformity.

The exact verdict is

$$(A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL).$$

There is no target divisor, prime or zero table, arithmetic Euler factor, root number, automorphy claim, unitary lift, or Hilbert–Pólya operator. Route B is unauthorized under `NO_BAD_EULER_OR_ROOT_NUMBER`.

6 Conclusion

C137 proves that nonlinear order sensitivity and trace-class determinant control persist on a compact parameter family. Connecting this source-owned family to any external target is a separate unsupported problem.